

Software Engineering

Name :-

Reg No :-

Topic :-

Ride Sharing

Purpose :-

An InDrive-style ride sharing app for Pakistan that connects passenger with riders. People can book affordable rides quickly and drivers can earn income using their own vehicles.

Ride-hailing reaches only about 5% of Pakistan population, compared to about 20% globally.

Users :-

- Passenger :- Request rides, pay and track trips.
- Drivers :- Accept rides and navigate to destination.
- Support Agents :- Check complaints, lost items.
- Verification Staff :- Check driver documents e.t.c.

Stakeholders :-

Everyone affected by or influencing the system, not just the users: platform owners and investors,

payment providers (JazzCash, Easy Paisa)
regulators (transport authorities and tax
tax authorities) competitors.

Five functional Requirements:-

- Passenger can choose pickup and drop-off points.
- Passengers can offer fare and drivers can accept.
- Drivers can accept or reject ride request.
- Passenger can track the ride live and share the trip.
- Passenger can pay by cash or wallet.

Five non functional Requirements:-

- 1) Performance :- Driver offers appear within 10sec.
- 2) Capacity :- Handles 20,000 concurrent active rides
- 3) Location Updates :- Driver location updates every 5sec
- 4) Compatibilities :- Works on low-end Android phones
- 5) Localization :- English and Urdu, fare in PKR.

Three Risks :-

- Driver Shortage :- A shortage of vehicles limit.
- Safety incidents :- Harassment, Accident, Fake.
- Price War :- Fare Bidding pushed fares so low that driver earn less.

Three Quality Attributes:-

- 1) Availability :- Stay online at peak hours.
- 2) Reliability :- Fare and driver location are accurate.
- 3) Security and Privacy :- Protects Phone no.

Where AI could Be used:-

- Driver Matching :- Picks the best nearby driver.
- ETA and Routes :- Predicts arrival time.
- Fare Suggestion :- Suggests a fair price.
- Demand forecasting :- Predicts busy areas.
- Safety detection :- Flags unusual routes.

What could gone wrong if AI is used incorrectly:-

- i) Matching if favour some driver.
- ii) ETA and routes if a wrong estimate
- iii) Fare Suggestion if too low-fare offered
- iv) Forecasting if wrong prediction is given.
- v) Safety detection if wrongly flags honest drivers.